

Oluwatobi Melvyn Mayungbo

Machine Learning and AI Engineer

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SUMMARY

Machine learning and AI engineer who builds from zero to production alone. Built sequence-based fraud detection served at 5.1 ms p99, a retrieval assistant with guardrails and golden-set evaluation, and an MCP server for AI assistants. Deploys with FastAPI and Docker on GCP Cloud Run, Railway and AWS.

SKILLS

Machine Learning: PyTorch, Deep Learning, Sequence Models (GRU, TCN, Transformers), LightGBM, Scikit-learn, XGBoost, Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

AI and LLM Systems: Retrieval-Augmented Generation (RAG), LLM Evaluation and Guardrails, Model Context Protocol (MCP), Hybrid Search (BM25, Dense Embeddings, RRF), Prompt Engineering, Citation Verification, Hugging Face, ONNX Runtime

Engineering and MLOps: Python, FastAPI, Docker, GCP Cloud Run, Railway, SQL, DuckDB, PostgreSQL, Redis, AWS ECS Fargate, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.
- Handled extreme class imbalance with focal loss and weighted sampling and isolated their failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling cost 0.042 PR-AUC. Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.

- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action